

Project Report: Airflow + dbt Analytics Pipeline - Data Engineer

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Family:	Data Engineer
Level:	Intermediate
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

Airflow DAG orchestrates dbt models on BigQuery/Snowflake with data quality tests.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Intermediate]

Chapter focus: Project Overview** *(Level: Intermediate)

Airflow + dbt Analytics Pipeline** is a intermediate-level end-to-end project for Data Engineer. **Duration:** 4-5 weeks. **Airflow DAG** orchestrates dbt models on BigQuery/Snowflake with data quality tests. **Tools:** Airflow, dbt, BigQuery/Snowflake.

Code Reference:

Objective: Airflow DAG with task dependencies
Objective: dbt models with tests
Objective: Data quality checks
Objective: Slack alert on failure

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

Data Engineer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a Data Engineer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: dbt [Intermediate]

Chapter focus: Phase: dbt** *(Level: Intermediate)

Staging + mart models. Complete these tasks: sources.yml; schema tests. Deliverable for this phase: dbt docs site.

Code Reference:

- [] sources.yml
- [] schema tests

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase dbt is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

Data Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the dbt docs site to the teacher.

Chapter 3: Phase: Airflow [Intermediate]

Chapter focus: Phase: Airflow** *(Level: Intermediate)

DAG scheduling. Complete these tasks: Sensors; Retries. Deliverable for this phase: DAG running daily.

Code Reference:

- [] Sensors
- [] Retries

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Airflow is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

Data Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the DAG running daily to the teacher.

Chapter 4: Phase: Quality [Intermediate]**Chapter focus: Phase: Quality** *(Level: Intermediate)**

dbt tests + custom. Complete these tasks: Unique; Not null; Freshness. Deliverable for this phase: Quality report.

Code Reference:

- [] Unique
- [] Not null
- [] Freshness

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase Quality is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

Data Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Quality report to the teacher.

Chapter 5: Phase: Alert [Intermediate]**Chapter focus: Phase: Alert** *(Level: Intermediate)**

Slack on failure. Complete these tasks: Webhook; Run summary. Deliverable for this phase: Alert screenshot.

Code Reference:

- [] Webhook
- [] Run summary

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Alert is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

Data Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Alert screenshot to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Intermediate]

Chapter focus: Final Deliverables and Review *(Level: Intermediate)**

Submit all final deliverables: dbt project; Airflow DAG; dbt docs URL; Quality report. Skills demonstrated: Airflow, dbt, Data quality. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

```
Deliverable: dbt project
Deliverable: Airflow DAG
Deliverable: dbt docs URL
Deliverable: Quality report
```

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide

What it actually is

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits dbt project and passes the rubric.